

1) Multimedia

1.1 Definition

The term multimedia is made up of 2 words “multi” and “media”. “Multi” refers to many while “media” refers to different types of communication forms. Hence, the term “multimedia” refers to different types of media.

Multimedia is the integration of **more than one type of media** (such as **text, images/graphics, audio, video, and animation**) in a single platform to communicate information effectively.

Multimedia Technology refers to the **tools and techniques** used to create multimedia content such as **apps, movies, games, websites, advertisements, and presentations**.

Example

An educational mobile app that shows **animated experiments**, gives **voice explanations**, includes **video lessons**, and provides **quizzes** is a multimedia system.

1.2 Apps/Platforms that use Multimedia (Examples)

- **Duolingo:** uses **text + voice + sound effects + animation** for language learning
- **TikTok:** uses **video + music + filters + text overlays**
- **YouTube:** uses **video + audio + thumbnails + live chat**
- **Google Earth:** uses **satellite images + text labels + 3D maps**

1.3 Components of Multimedia (5 Main)

1. **Text**
2. **Graphics**
3. **Audio**
4. **Video**
5. **Animation**

1.4 Applications of Multimedia

a) Education

Definition: Use of multimedia to improve learning using videos, animations, simulations, and interactive content.

Example: e-books, online courses, science simulations, learning apps.

b) Entertainment

Definition: Use of multimedia to provide enjoyment through interactive media.

Example: video games, movies, music, virtual reality.

c) Business

Definition: Multimedia used for marketing and promotion to attract customers.

Example: advertisements, product promos, digital posters, brand videos.

d) Communication

Definition: Multimedia makes communication effective using audio/video and visuals.

Example: video conferencing, social media posts, messaging apps using stickers/videos.

e) Content Creation

Definition: Multimedia tools used to create engaging digital content for online platforms.

Example: creating reels/short videos, posters, animations, voiceovers for social media.

2) Text (Multimedia Component)

2.1 Definition

Text refers to the **characters displayed on screen**, also called **alphanumeric values**.

A meaningful combination of characters forms information for readers.

Example

Headings, captions, subtitles, labels in apps and videos are text elements.

2.2 Types of Text Editors (Examples)

Text editors are software used to create and edit text.

- **Notepad**
- **WordPad**
- **Microsoft Word**

2.3 Text File Formats (Important)

a) TXT

Definition: A standard text file format that stores **only plain text** (no formatting).

Example: notes.txt

b) RTF (Rich Text Format)

Definition: Stores text along with formatting like **font size, colour, line spacing, indentation** etc.

Example: letter.rtf

c) DOCX

Definition: Microsoft Word document format that stores text with formatting and advanced features.

Example: report.docx

3) Graphics (Multimedia Component)

3.1 Definition

Graphics are **visual elements** seen on screen, including **images, pictures, shapes, and drawings**, which help present information visually.

Example

Photos in a website, icons in an app, diagrams in a book are graphics.

3.2 Graphics Editors (Examples)

- **Pixlr X**
- **Adobe Photoshop**
- **Adobe Illustrator**

3.3 Raster and Vector Graphics

3.3.1 Raster Graphics (Bitmap Graphics)

Definition: Raster graphics are images made up of **tiny dots** called **pixels**. Each pixel has a specific color, and when you put all these pixels together, they form a complete picture. When zoomed in too much, raster images become **blurred/pixelated**. The Raster format of graphics is also known as Bitmap Graphics.

Pixel (Very Important)

Pixel stands for **Picture Element** and is the **smallest unit** of an image displayed on a screen.

Example

A photograph taken from a phone camera is usually raster.

Common Raster File Formats

PNG (Portable Network Graphics)

- Supports **transparency**
- Good for logos and images with transparent background
Example: logo.png

GIF (Graphics Interchange Format)

- Supports **animation**

- Used for short looping animations
Example: funny.gif

JPG/JPEG (Joint Photographic Experts Group)

- High quality with **good compression**
- Smaller file size, most common for photos
Example: photo.jpg

3.3.2 Vector Graphics

Definition: Vector graphics are made using **Bezier curves** stored as **mathematical formulas**. They can be resized without losing quality.

Example

Company logos and icons designed in Illustrator are often vector.

Common Vector File Format

SVG (Scalable Vector Graphics)

- Most common vector format
- Supported by many applications
Example: icon.svg

3.3.3 PDF File Format

PDF (Portable Document Format) is a versatile format used for sharing documents. It can contain **both raster and vector** graphics depending on how it is saved.

Example

A scanned document (raster) inside a PDF or a digitally exported report (vector + text) in PDF.

4) Graphics Editing (Pixlr X)

4.1 Pixlr X (Definition)

Pixlr X is a **free online graphics/photo editor** used for cropping, resizing, adding text, shapes, filters, and layers.

4.2 Pixel Art (Topic from Book)

Pixel Art is a digital art style where images are created using **visible pixels** like in old-school video games.

Examples of Pixel Art Apps

- **Piskel**
- **Aseprite**
- **Pixilart**
- (and other pixel art editors)

4.3 Main Areas of Pixlr X Interface

1. **Canvas area** — workspace to view/edit image
2. **Tools panel** — editing tools and features
3. **Layers panel** — manage image/text/shape layers
4. **Navigation tools** — zoom, pan, rotate

4.4 Steps: Creating a New Project in Pixlr X

1. Go to Pixlr website and open **Pixlr Express** or **Pixlr Editor**.
2. Choose **Open existing image** or **Create new file**.
3. Click **Create new** for a blank project.
4. Select a preset size or enter custom width/height (e.g., **1920 × 1080**).
5. Enter project name (optional).
6. Click **Create**.
7. Start editing your project.

4.5 Steps: Working with Text in Pixlr X

1. Create/open a project.
2. Click **Text tool (T)** on left panel.
3. Click **Add new text**.
4. Type your text (or paste).
5. Click **Arrange** to edit text options.
6. Change font, size, style, colour, alignment, spacing.
7. Resize using corner handles; move by dragging or arrow keys.

8. Rotate using the circular handle on top.
9. Manage text from **Layers panel** (hide/lock/duplicate/delete/edit later).

4.6 Steps: Working with Shapes in Pixlr X

1. Click **Arrange** tool.
2. Choose **Shape**.
3. A default square appears.
4. Select shape type (circle, etc.) from the **Type** section.
5. Edit fill colour, outline colour, shadow, etc.
6. Resize using corner handles.
7. Rotate using circular rotation handle.

4.7 Image Placement — Inserting and Editing (Basic)

- Insert an image into your design and adjust size, position, rotate, and apply edits like crop/filters.

4.8 Steps: Working with Layers in Pixlr X

1. Open **Layers panel** on right side.
2. Click + to create a new layer.
3. Select the type of layer you want (text/shape/image, etc.).
4. Click the **three dots** on a layer for more options.
5. Rename layer if needed.
6. Lock or hide layer.
7. Adjust transparency and blend mode.
8. Rearrange layer order by drag-and-drop.

4.9 Steps: Inserting Image in Existing Project (Pixlr X)

1. Click the **Element** icon (bottom left).
2. Click **Add Media**.
3. Select the image file from your computer.
4. Click **Open**.
5. Image appears on your canvas (resize/move as required).

4.10 Steps: Using Cutout Tool (Pixlr X)

1. Open an image in Pixlr X.
2. Click **Cutout** tool from left panel.
3. Choose cutout type/shape and set **softness** for edge smoothness.
4. Drag over the object you want to cut out.
5. Use **Keep** and **Remove** to refine selection.
6. Use **Draw cutout** to freely add/remove parts.
7. Use **Lasso cutout** to draw selection shapes for add/remove.

8. Finalize the cutout result.

4.11 Steps: Exporting Project (Pixlr X)

1. Click **Save** (bottom right).
2. Choose file name, format, and quality.
3. Select format: **JPG / PNG / WEBP / PXZ**
 - JPG: common for web/print
 - PNG: transparency
 - WEBP: better compression
 - PXZ: keeps layers for future editing
4. Adjust quality slider (higher = better quality but larger file size).
5. Click **Save As**.
6. Choose location and save file.

5) Audio (Multimedia Component)

5.1 Definition

Audio refers to **sound** in multimedia such as music, speech, instruments, and sound effects.

5.2 How audio is produced/stored in computers

Sound is produced due to **vibrations**. In computers, these vibrations are converted into **digital data** and saved as an **audio file**.

5.3 Audio Waveform

Definition: An **audio waveform** is a graph that represents how sound changes over time.

Two Important Terms

- **Amplitude:** shows **loudness** of sound (height of waveform)
- **Wavelength:** shows vibration pattern over time (width/spacing)

5.4 Common Audio File Formats

MP3 (MPEG Audio Layer 3)

- Most common
- High compression → smaller size
Example: song.mp3

AAC (Advanced Audio Coding)

- Improved compression compared to MP3
Example: audio.aac

WAV (Waveform Audio File Format)

- Uncompressed → large size but best original quality
Example: recording.wav

6) Audio Editing (Audiomass)

6.1 Audiomass (Definition)

Audiomass is a free online audio editor used for recording, cutting, applying effects, and exporting audio.

6.2 Main Areas of Audiomass

1. **Menu Bar** — file, edit, view, effects, help
2. **Waveform Display** — shows waveform and supports cut/copy/paste/trim/delete
3. **Control Panel** — play, pause, stop, record, volume adjustment

6.3 Steps: Recording Audio in Audiomass

1. Open the Audiomass website in a browser.
2. Click **Record** (or press **R**).
3. A new recording dialog appears.
4. Select correct microphone under **Devices**.
5. Click **Start Recording**.
6. Record your voice/sound.
7. Click **Finish Recording**.
8. Click **Open Recording** to load it into the waveform editor.

6.4 Steps: Importing Audio in Audiomass

1. Click **File** (top left).
2. Select **Load from Computer**.
3. Choose your audio file and click **Open**.
4. Audio appears as waveform.

6.5 Steps: Cutting Audio Waveform

1. Click and drag to select unwanted part.
2. A selection box appears.
3. Click **Scissors** icon to cut selected part.

6.6 Steps: Applying Effects

1. Select the part of audio.
2. Click **Effects** from menu bar.
3. Choose effect (normalize, fade in/out, reverse, pitch shift, reverb, etc.).
4. Adjust parameters if needed.
5. Click **Apply**.
6. Play audio to check result.

6.7 Steps: Exporting Audio

1. Click **File**.
2. Choose **Export/Download**.
3. Name your file.
4. Choose format **MP3 or WAV**.

5. Select quality/bitrate and channels.
6. Click **Export**.

7) Video (Multimedia Component)

7.1 Definition

Video is a collection of pictures called **frames** that play continuously to create the illusion of motion. It also contains an audio file that plays along with the frames.

7.2 Frame Rate (FPS)

The series of frames that runs or plays in a certain time interval is known as Frame Rate which can be measured as frames per second (FPS). The amount of frames per second determines whether a video is smooth or choppy.

- Higher FPS → smoother video
 - Lower FPS → choppy video
- Common: **24fps, 25fps, 50fps, 60fps**

7.3 Resolution

Resolution is the number of pixels in an image/video, written as **width × height** (e.g., 1920×1080). Higher resolution → clearer quality but larger file size.

Common Resolutions

- **SD (Standard Definition):** 720×480 Pixels. This is a low resolution that is used for older TVs and DVDs. It can show basic colors and shapes, but not much detail.
- **HD (High Definition):** 1280×720 pixels. This is a medium resolution that is used for newer TVs and online videos. It can show more details and colors than SD, but not as much as FHD.
- **FHD (Full High Definition):** 1920×1080 pixels. This is a high resolution that is used for newer TVs, monitors, and laptops. It can show a lot of details and colors, and it is the most common resolution for HD videos and games.

7.4 Video File Formats (Important)

MP4 (MPEG-4 Part 14)

- This is the most commonly used video format due to its high level of compression capabilities
Example: project.mp4

MOV (QuickTime Movie)

- It is commonly used to store videos which may be compressed or uncompressed as per the requirement.
Example: clip.mov

AVI (Audio Video Interleave)

- This video format is also commonly used to store videos which may be compressed or uncompressed as per the requirement.
Example: video.avi

8) Video Editing (CapCut)

8.1 CapCut (Definition)

CapCut is a free online/mobile video editing tool used to cut clips, arrange them, add transitions, effects, text, and export videos.

8.2 Main Areas of CapCut

1. **Timeline Area** — arrange clips, audio, effects, text
2. **Tools Panel** — access media, audio, transitions, effects, text options
3. **Preview Area** — preview the edited video
4. **Export & Share Options** — export/download and share video

8.3 Steps: Creating New Project in CapCut

1. Open CapCut website.
2. Click **Create new video**.
3. Choose aspect ratio (e.g., **16:9**).
4. New project opens in editing interface.
5. Rename project title if needed.

8.4 Steps: Importing Videos

1. Click **Media** from left panel.
2. Click **Upload**.

3. Select **From this device**.
4. Choose video/media files and click **Open**.
5. Files appear in Project section.

8.5 Steps: Working in Timeline

1. Click a media file to add it to timeline **or** drag and drop into timeline.
2. Clips appear in the order you place them.
3. Rearrange clips by dragging to new position.
4. Zoom in/out timeline using zoom slider.
5. Press **Spacebar** or play button to preview.
6. Drag cursor/playhead to jump to any time.
7. Current time shows in preview area.

8.6 Steps: Trimming and Cutting Videos

Trimming

1. Select clip in timeline.
2. Drag start/end handles to adjust duration.

Cutting (Split)

1. Move playhead to cut point.
2. Click **Split**.
3. Select unwanted part and click **Delete**.

8.7 Steps: Adding Transitions

1. Hover between two clips.
2. Click **Add Transition** button.
3. Choose transition from transition menu.
4. Adjust duration by dragging slider edges.
5. Repeat for other clips.

8.8 Steps: Adding Effects

1. Click **Effects** from left panel.
2. Choose category and effect.
3. Place playhead where effect should start.
4. Click effect to apply it to timeline.
5. Double click effect layer to open settings.
6. Adjust properties.
7. Repeat if needed.

8.9 Steps: Adding Text Effects

1. Click **Text** tool.
2. Choose basic text or a text template.
3. Click **Add heading** (example) → text box appears.
4. Double click text box → type text.
5. Edit text properties from properties panel.
6. Adjust text duration by dragging text clip edges.

8.10 Steps: Exporting Project (CapCut)

1. Open your project.
2. Click **Export** (top right).
3. Click **Download**.
4. Select resolution, quality, and frame rate.
5. Click **Export**.
6. After export completes, click **Download** to save.

9) Animation

9.1 Definition

Animation is the movement of pictures or graphics. In multimedia, it usually refers to **computer-generated motion** used in cartoons, games, advertisements, and films.

Example

Cartoon character movement or animated text in ads.

9.2 Types of Animation

2D Animation

Definition: Flat animation with only **width and height**. Usually drawn **frame by frame** in animation software.

Example: 2D cartoons, simple educational animations.

3D Animation

Definition: Animation with **width, height, and depth** (perspective). Usually modelled and rigged before movement.

Example: 3D movies, modern game characters, 3D animated ads.